

PANDA Component Description: *pwm_port*

Overview

The *pwm_port* component implements a simple pulsewidth modulated output port with adjustable resolution and modulation frequency.

Generics

Name	Type	Default	Description
<i>clk_frequency</i>	integer	10'000'000	Clock frequency in Hz
<i>pwm_frequency</i>	integer	10'000	Frequency of the PWM output signal in Hz
<i>pwm_res_bits</i>	integer	8	Resolution of the PWM output in bits (1...8)

External Ports

Name	Type	Width	Description
<i>pwm</i>	<i>std_logic</i>	1	PWM output signal

Registers

Value Register

Name: VALUE

Address: Base + 0x00

Name	VALUE
R/W	R/W
Reset Value	0x00

The *VALUE* register holds the value that is pulsewidth modulated. The width of *VALUE* is defined by *pwm_res_bits*. *VALUE* is right adjusted. Unused higher order bits are always read as '0' and can be written with any value.

Calculations

The exact frequency of the PWM-output can be calculated with the following equation:

$$\text{exact frequency} = \frac{\text{clk_frequency}}{(2^{\text{pwm_res_bits}} - 1) \cdot \text{prescaler}}$$

where *prescaler* is defined as the **rounded** result of the following equation:

$$\text{prescaler} = \frac{\text{clk_frequency}}{(2^{\text{pwm_res_bits}} - 1) \cdot \text{pwm_frequency}}$$